A Non-Split Regge–Wheeler Self-Extension and a Defective Schwarzschild Resonance in Pure Weyl Gravity

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Abstract

We determine the differential-module and resonance structure of axial Schwarzschild perturbations in four-dimensional pure Weyl gravity. The six-state Bach system has an exact filtration with successive quotients given by two spin-two Regge–Wheeler modules and one spin-one Maxwell module. The repeated spin-two block is a non-split self-extension and is the first jet of a four-state spin-two/spin-one family.

After Liouville reduction, the extension is represented by a scalar projective cocycle modulo the symmetric-square operator

$$\mathcal{K}_U = D^3 + 4UD + 2D(U).$$

For axial $\ell = 2$ we prove the exact normal form

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

The class in brackets is exactly the transverse-traceless critical spin-two mass jet. Thus the local parameter relation is $m = (i\omega/2)\tau$, modulo rational triangular gauge. The required gauge grows linearly at infinity, precisely matching the derivative of the moving massive Jost phase. The long-range Coulomb tail cancels the apparent logarithmic phase derivative, so the gauge preserves the differentiated Jost classes modulo scalar normalization.

At the covariant parent-operator level, the pure-Weyl metric Green function is exactly the first mass derivative of the Einstein–Weyl spin-two resolvent. This determines both leading Laurent coefficients and the isolated-resonance time contribution directly from the massive QNM velocity and projector derivative.

At a simple spin-two quasinormal frequency the complete connection admits two possible local Smith types. A validated projective Evans calculation selects the defective type $(0, 0, 2)$ at one damped Schwarzschild mode. The geometric root vector is Einstein, whereas its generalized root vector has nonzero projection to the Ricci-carrier quotient. We further show that the compact-source, compact-observation outgoing Green operator on the Schwarzschild exterior has a nonzero rank-one pole of order two at this frequency. The same selector is the nonzero critical-mass velocity of the massive spin-two QNM. The resulting transverse confluence has projectors of size m^{-1}, positive branch metrics of condition number m^{-2}, a local contour limit with a Jordan polynomial, and a hyperbolic renormalized Krein form.

Promotion to a global physical resolvent and to a full retarded generalized ringdown expansion requires a separate analytic Fredholm realization, weighted endpoint domains, and bounded metric reconstruction.

1 Introduction and main results

Pure Weyl gravity has a fourth-order metric equation, but its Schwarzschild perturbations are not best understood as an unstructured fourth-order system. In the axial $\ell = 2$ sector the six-dimensional radial system has the filtered form

$$0 \subset M_2 \subset E_2 \subset M_{\mathrm{Bach}}, \qquad \mathrm{gr}\, M_{\mathrm{Bach}} \simeq M_2 \oplus M_2 \oplus M_1,$$

where M_2 is the spin-two Regge–Wheeler module and M_1 is the spin-one Maxwell module. The middle object E_2 is a self-extension

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0.$$

The decisive point is that this extension is non-split over the rational differential field. The two spin-two layers therefore share scalar characteristics without defining two rationally separable radial branches.

This paper establishes six central results.

1. The repeated spin-two block is a non-split Regge–Wheeler self-extension, realized exactly as a parameter jet.

2. Its projective extension class has the elementary normal form

$$[\mathcal{I}_{\mathrm{Bach}}] = \frac{i\omega}{2}\left[1 - \frac{2}{r}\right].$$

3. The class $[1 - 2/r]$ is exactly the transverse-traceless critical mass jet. Its linearly growing equivalence gauge is endpoint-compatible: the Coulomb tail cancels the apparent logarithmic derivative of the massive Jost phase.

4. At a certified damped Schwarzschild quasinormal frequency the connection is defective. The associated generalized root vector has a nonzero carrier quotient, and the exterior cut-off outgoing Green operator has a genuine rank-one double pole.

5. The selector equals the normalized critical-mass QNM velocity and is nonzero. The finite-mass pair is a transverse unfolding of the EP2; its singular projectors, local two-pole contour, positive metrics, and Krein form have exact confluent limits.

6. The covariant pure-Weyl metric Green function is the mass derivative of the massive spin-two resolvent. Its double coefficient is controlled by QNM motion, its simple coefficient by projector motion, and its local polynomial signal obeys an exact rank-one excitation/observation rule.

The third result is stronger than a statement about a finite connection matrix. It is a meromorphic differential-response theorem for compact radial sources and local radial observations. It is deliberately weaker than a causal spacetime resolvent theorem: no inverse-Laplace deformation, late-time expansion, or $te^{i\omega_n t}$ assertion is made here.

Logical status

Result	Required input	Status
RW/RW/Maxwell filtration and partial jet	Exact rational identities and reconstruction rank	Exact
Non-split spin-two extension	Exhaustive rational coboundary obstruction	Exact for real $\omega > 0$
Cocycle and local critical-mass normal form	Exact substitution in $\mathbb{C}(r, \omega)$	Exact for $\omega \neq 0$
Endpoint-compatible mass comparison	Moving massive/Coulomb Jost germs and analytic normalization	Exact at the simple QNM
Defective physical specialization	Winding-one contour, local selector, spin-one unit	Validated theorem
Non-Einstein generalized root	Defective Smith type and exact filtration	Exact consequence
Exterior cut-off outgoing Green double pole	Analytic Jost maps, transparent cuts, finite-interval reduction	Exact/validated theorem
Causal spacetime pole and generalized ringdown	Global weighted Fredholm and contour theory	Not claimed
Critical-mass unfolding and QNM velocity	Endpoint-compatible jet and nonzero certified selector	Exact/validated theorem
Parent mass-derivative Green identity	Fixed-domain parent inverse and analytic massive QNM branch	Exact operator theorem
Local two-pole contour limit	Analytic normal form and residue calculus	Exact local theorem

2 Differential-module conventions

We work over

$$K = \mathbb{C}(r, \omega), \qquad D = f\partial_r, \qquad f = 1 - \frac{2}{r},$$

with $\omega \neq 0$ unless a threshold limit is stated explicitly. All rational gauges and module morphisms are over the differential field (K, D). We use $D(q)$, $D(U)$, and D^3q, rather than primes, to avoid confusing the tortoise derivative with ∂_r.

The axial spin-two and spin-one scalar operators are

$$L_2 = D^2 + U_2, \qquad U_2 = \omega^2 - f\left(\frac{6}{r^2} - \frac{6}{r^3}\right),$$

$$L_1 = D^2 + U_1, \qquad U_1 = \omega^2 - f\frac{6}{r^2}.$$

The latter is the $\ell = 2$ Maxwell Regge–Wheeler equation. A rank-two scalar module is represented by its companion system on $(y, Dy)^T$.

The source Einstein layer is the kernel of the linearized Ricci/Schouten carrier map. The target spin-two quotient is its physical Regge–Wheeler carrier. The third quotient is the Maxwell class of the target Einstein gauge vector. Consequently, "non-Einstein" below means nonzero projection to the gauge-invariant carrier quotient, not failure of a coordinate gauge condition.

3 Exact axial RW/RW/Maxwell filtration

In the certified factor gauge the six-state system is

$$\begin{aligned} DX &= AX + EY + CZ, \\ DY &= AY + DZ, \\ DZ &= A_1 Z, \end{aligned} \tag{1}$$

where $X, Y, Z \in K^2$, the companion matrices A and A_1 represent L_2 and L_1, and the forcing blocks factor through a single scalar source:

$$E = USJ, \qquad C = USN.$$

The rational reconstruction has full row rank on the declared domain.

Theorem 3.1 (Filtered axial Bach module)**.** *For axial $\ell = 2$ Schwarzschild perturbations, the reduced six-state Bach module has a filtration whose successive simple factors are*

$$M_2, \qquad M_2, \qquad M_1.$$

The middle two-spin-two block is an extension E_2 of M_2 by M_2. The quotient M_1 is the Maxwell module of the carrier gauge vector. The filtration is exact but is not asserted to be a direct sum.

The theorem is an exact rational statement. It is independent of endpoint normalization, scattering conventions, and the later QNM specialization.

4 Partial-jet realization

Introduce the four-state family

$$\mathcal{B}(\tau) = \begin{pmatrix} A + \tau E & D + \tau C \\ 0 & A_1 \end{pmatrix}.$$

Let its fundamental matrix be

$$\Phi_4(\tau) = \begin{pmatrix} P(\tau) & Q(\tau) \\ 0 & R \end{pmatrix}.$$

Theorem 4.1 (Exact partial jet)**.** *The six-state generator in (1) is the first spin-two-row jet of $\mathcal{B}(\tau)$ at $\tau = 0$. Its fundamental matrix is*

$$\Phi_6 = \begin{pmatrix} P & \dot{P} & \dot{Q} \\ 0 & P & Q \\ 0 & 0 & R \end{pmatrix}_{\tau=0}, \qquad \det \Phi_6 = (\det P)^2 \det R.$$

Consequently any factor-adapted endpoint connection has the triangular form

$$T(\omega) = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix}, \qquad \det T = a^2 g.$$

Proof. Differentiate $D\Phi_4(\tau) = \mathcal{B}(\tau)\Phi_4(\tau)$ at $\tau = 0$. The equations for $P, \dot{P}, Q, \dot{Q}, R$ reproduce the three rows of (1) column by column. The determinant identity follows from block triangularity. The same identity applied to endpoint transition matrices gives the displayed connection form. □

The parameter τ is intrinsic at this stage. Identifying it with a physical Fierz–Pauli mass requires an additional finite-mass reduction and compatible endpoint germs; Section 15 records the exact remaining gate.

5 Projective gauge and symmetric-square cohomology

Consider a scalar self-extension

$$Lx = (s_1 D + s_0)y, \qquad Ly = 0, \qquad L = D^2 + U. \tag{2}$$

Set

$$\mathcal{I} = s_0 - \frac{1}{2}D(s_1), \qquad \mathcal{K}_U = D^3 + 4UD + 2D(U).$$

Lemma 5.1 (Explicit triangular gauge)**.** *For $q \in K$, define*

$$Q(q) = qD - \frac{1}{2}D(q).$$

On every solution $Ly = 0$,

$$[L, Q(q)]y = -\frac{1}{2}\mathcal{K}_U(q)y.$$

Thus the change of lift $x \mapsto x + Q(q)y$ changes the scalar extension density by

$$\mathcal{I} \longmapsto \mathcal{I} - \frac{1}{2}\mathcal{K}_U(q).$$

Proof. Expand $LQ(q) - Q(q)L$, use the Leibniz rule for D, and then set $D^2y = -Uy$. The coefficient of Dy cancels, while the coefficient of y is $-\frac{1}{2}[D^3q + 4UDq + 2D(U)q]$. □

The extension is therefore represented by

$$[\mathcal{I}] \in K/\mathcal{K}_U K.$$

Because Lemma 5.1 contains the factor 1/2, removal of a term $\mathcal{K}_U q$ uses the triangular gauge $Q(2q)$. This factor does not alter the cohomology quotient but matters in an explicit frame calculation.

The equation $\mathcal{K}_U q = 0$ is the symmetric-square equation of L. If y_1, y_2 span the scalar solution space, then its solution space is spanned by $y_1^2, y_1 y_2, y_2^2$. The same operator controls both the projective obstruction and trace-free horizontal endomorphisms.

6 Mass-direction normal form of the Bach cocycle

For the axial $\ell = 2$ Bach extension, rational reduction gives

$$\mathcal{I}_{\text{Bach}} = \frac{i(r-2)(2r\omega^2 + 3\omega^2 + 12)}{5r^4\omega}.$$

Theorem 6.1 (Mass-direction normal form)**.** *Let*

$$q(r,\omega) = -\frac{i}{120\omega}\left(15r + 13 + \frac{12}{r} + \frac{9}{r^2}\right).$$

For $\omega \neq 0$,

$$\mathcal{I}_{\text{Bach}} = \mathcal{K}_{U_2} q + \frac{i\omega}{2} f.$$

Consequently

$$[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[f] \quad \textit{in} \quad K/\mathcal{K}_{U_2}K.$$

In particular, $[f] \neq 0$ wherever the Bach self-extension is non-split.

Proof. Apply $D = f\partial_r$ three times to q and substitute

$$U_2 = \omega^2 - f\left(\frac{6}{r^2} - \frac{6}{r^3}\right)$$

in $\mathcal{K}_{U_2}q = D^3q + 4U_2Dq + 2D(U_2)q$. Clearing the common denominator $120r^4\omega$ reduces the claim to a polynomial identity in r and ω. Exact coefficient comparison gives the displayed formula. The quotient identity follows immediately. If $[f] = 0$, then $[\mathcal{I}_{\mathrm{Bach}}] = 0$, which would split the extension. □

Proposition 6.2 (Static exactness and the threshold lattice)**.** *Write*

$$q(r,\omega) = \frac{q_{-1}(r)}{\omega}, \qquad q_{-1}(r) = -\frac{i}{120}\left(15r + 13 + \frac{12}{r} + \frac{9}{r^2}\right),$$

and $U_2(\omega) = U_{2,0} + \omega^2$. *Then*

$$\mathcal{K}_{U_2} = \mathcal{K}_{U_{2,0}} + 4\omega^2 D$$

and

$$\boxed{\mathcal{I}_{\mathrm{Bach}} = \frac{1}{\omega}\mathcal{K}_{U_{2,0}}q_{-1} + \omega\left(4Dq_{-1} + \frac{i}{2}f\right).}$$

In particular,

$$\operatorname*{Res}_{\omega=0}\mathcal{I}_{\mathrm{Bach}} = \mathcal{K}_{U_{2,0}}q_{-1}.$$

Thus the entire $1/\omega$ *part is already exact for the static symmetric-square operator. If the projective cokernels are identified continuously at threshold, then*

$$\boxed{\lim_{\omega\to 0}\frac{[\mathcal{I}_{\mathrm{Bach}}]}{\omega} = \frac{i}{2}[f].}$$

Remark 6.3 (Bulk class versus spectral frame)*.* The intrinsic extension class vanishes linearly at threshold in the meromorphic projective frame. The gauge producing this representative is itself $O(\omega^{-1})$, so it is singular in a holomorphic physical Jost frame. The threshold issue is therefore a spectral-frame or lattice problem, not a divergence of the bulk projective class. This proposition does not construct a threshold-uniform splitting or justify commuting the limits $\omega \to 0$ and $\tau \to 0$.

Remark 6.4 (Mass-shaped is not yet physical mass)*.* The representative $f = 1 - 2/r$ is the radial direction expected from a mass term after a helicity-two reduction. The theorem proves a local cohomology normal form, not equality with a particular Einstein–Weyl mass parameter. If a finite-mass reduction gives

$$\mathcal{I}_{\mathrm{mass}} = c(\omega)f + \mathcal{K}_{U_2}p,$$

then the two extensions are locally equivalent precisely when $c(\omega) \neq 0$; their parameter normalization is $2c(\omega)/(i\omega)$.

7 Period formula for the connection shear

Let y_1, y_2 be a scalar basis and set

$$Y = \begin{pmatrix} y_1 & y_2 \\ Dy_1 & Dy_2 \end{pmatrix}, \qquad W = y_1Dy_2 - y_2Dy_1.$$

Because the companion matrix has zero trace, $D(W) = 0$.

Theorem 7.1 (Symmetric-square period matrix)**.** *For the family*

$$(D^2 + U + \tau V)y = 0,$$

the interaction-picture variation satisfies

$$Y^{-1}\partial_\tau A\, Y = \frac{V}{W}\begin{pmatrix} y_1y_2 & y_2^2 \\ -y_1^2 & -y_1y_2 \end{pmatrix}.$$

Consequently the spin-two connection shear is determined by the three periods

$$\int \frac{V}{W} y_1^2\, dr_*, \qquad \int \frac{V}{W} y_1y_2\, dr_*, \qquad \int \frac{V}{W} y_2^2\, dr_*.$$

Proof. In the basis $(y, Dy)^T$,

$$A(\tau) = \begin{pmatrix} 0 & 1 \\ -(U+\tau V) & 0 \end{pmatrix}, \qquad \partial_\tau A = \begin{pmatrix} 0 & 0 \\ -V & 0 \end{pmatrix}.$$

Insert

$$Y^{-1} = \frac{1}{W}\begin{pmatrix} Dy_2 & -y_2 \\ -Dy_1 & y_1 \end{pmatrix}$$

and multiply. Integration of the interaction-picture equation gives the three entries. □

A coboundary $\mathcal{K}_U q$ changes these periods by endpoint terms. This is the connection-level form of Lemma 5.1: rational triangular gauges alter Jost frames but not the intrinsic divisor or Smith selector once the endpoint units are tracked.

8 Nonsplitting, endomorphisms, and Fitting ideals

Theorem 8.1 (Non-split physical-axis extension)**.** *For every real $\omega > 0$, the axial $\ell = 2$ spin-two block*

$$0 \longrightarrow M_2 \longrightarrow E_2 \longrightarrow M_2 \longrightarrow 0$$

is non-split over $\mathbb{C}(r)[D]$.

Certificate-supported exact proof. The rational splitting equation reduces, after exhaustive pole and degree bounds, to a finite coefficient system. Fixed minors of its coefficient and augmented matrices are

$$\det M_{[0,1,2]} = 3456\,\omega^{10}, \qquad \det[M \mid \mathrm{rhs}]_{[0,1,2,5]} = -645120\, i\,\omega^9.$$

They are simultaneously nonzero for every real $\omega > 0$, so the inhomogeneous coboundary equation is inconsistent. The fixed-minor argument also covers the former frame event $\omega^2 = 3$. □

The scalar Regge–Wheeler module is simple on the positive real axis and has scalar rational endomorphism ring. The only nonzero complex reducibility points allowed by the terminal Riccati ansatz with matching horizon and infinity signs are the algebraically special frequencies

$$\omega = \pm i\frac{(\ell-1)\ell(\ell+1)(\ell+2)}{12}.$$

For $\ell = 2$, these are $\omega = \pm 2i$, away from the physical axis.

Theorem 8.2 (Local commutant)**.** *For real $\omega > 0$,*

$$\mathrm{End}_{\mathbb{C}(r)[D]}(E_2) \simeq \mathbb{C}[\varepsilon]/(\varepsilon^2).$$

Every endomorphism is $aI + bN$, where $N^2 = 0$. Hence the only idempotents are $0, I$, the only involutions are $\pm I$, and every diagonalizable local endomorphism is scalar.

Proof. The submodule M_2 is the unique simple submodule: a second simple submodule mapping nontrivially to the quotient would split the extension. An endomorphism therefore induces scalars a, d on the submodule and quotient. Naturality of the nonzero extension class gives $a = d$. Subtracting aI leaves a map that vanishes on the submodule and quotient, so it factors through the quotient back into the submodule and is a scalar multiple of the canonical nilpotent N. The corollaries follow from $(aI + bN)^k = a^k I + ka^{k-1} bN$. □

For the triangular connection

$$T = \begin{pmatrix} a & b & c \\ 0 & a & d \\ 0 & 0 & g \end{pmatrix},$$

the Fitting ideals are

$$I_1 = (a, b, c, d, g), \qquad I_2 = (a^2, ad, bd - ac, ag, bg), \qquad I_3 = (a^2 g).$$

They give a normalization-independent local classification of all specializations.

9 Analytic Jost connections and Smith dichotomy

The moving Frobenius/Jost phases define analytic endpoint-selected planes on the certified spectral domains. Transport to a common ordinary radius and change to a factor-adapted frame. At a simple spin-two QNM ω_n, assume the spin-one coefficient $g(\omega_n) \neq 0$. Analytic row and column operations remove the spin-one row and column, leaving

$$T_2(\omega) = \begin{pmatrix} a(\omega) & b(\omega) \\ 0 & a(\omega) \end{pmatrix}, \qquad a(\omega_n) = 0, \quad a'(\omega_n) \neq 0.$$

Proposition 9.1 (Local Smith dichotomy)**.** *At ω_n,*

$$\mathrm{Smith}(T) = \begin{cases} (0,0,2), & b(\omega_n) \neq 0, \\ (0,1,1), & b(\omega_n) = 0. \end{cases}$$

The first case has geometric multiplicity one and one length-two root chain; the second has two independent length-one roots.

Proof. Let $z = \omega - \omega_n$. Since $a = a_1 z + O(z^2)$ with $a_1 \neq 0$, the determinant valuation of the two-spin-two block is two. If $b(0) \neq 0$, its first Fitting ideal is the unit ideal, so its exponents are $(0, 2)$. If $b(0) = 0$, all entries vanish to first order and the exponents are $(1, 1)$. Appending the spin-one unit adds a zero exponent. □

Projectively, write the selected endpoint lines in a common chart as

$$v_H(\omega, \tau), \qquad v_\infty(\omega, \tau), \qquad \delta = v_\infty - v_H.$$

At a simple root,

$$\frac{b(\omega_n)}{a'(\omega_n)} = \frac{\delta_\tau(\omega_n, 0)}{\delta_\omega(\omega_n, 0)}.$$

This ratio is invariant under analytic nonzero endpoint units and under moving the matching section.

10 Certified QNM specialization

The validated contour is oriented positively and encloses the disk

$$\Omega_n = \left\{\omega : \left|\omega - \left(-0.3736716844180418 + 0.0889623156889357\, i\right)\right| \le 10^{-7}\right\}.$$

Panelwise projective enclosures exclude zero on the contour, include all chart-transition units, and give winding number +1. Thus the disk contains exactly one simple spin-two zero.

On the localized disk, the intrinsic tangent is enclosed by

$$\delta_\tau(\Omega_n) \subset [0.0010932 \pm 7.60 \times 10^{-8}] + i[-0.0002195 \pm 8.89 \times 10^{-8}],$$

which excludes zero. An independent local calculation gives a spin-one mismatch modulus greater than 1/2500.

Theorem 10.1 (Certified defective Schwarzschild resonance)**.** *The complete axial connection has Smith valuations*

$$(0, 0, 2)$$

at the unique QNM in Ω_n*. The algebraic multiplicity of its spin-two divisor is two, its geometric multiplicity is one, and it has one length-two connection root chain.*

Proof. The winding certificate gives one simple spin-two zero. The local tangent enclosure proves $b(\omega_n) \neq 0$, while the spin-one enclosure proves $g(\omega_n) \neq 0$. Proposition 9.1 applies. □

11 Resonant evaluation of the self-extension

The projective selector, adjoint overlap, root-chain coefficient, and Green-pole residue are different representations of one resonant functional. This is the finite-interval Evans/Melnikov mechanism; compare the general relation between Evans derivatives and adjoint generalized eigenfunctions in [1].

Let

$$L(\omega) : X \longrightarrow Y$$

denote the analytic finite-interval spin-two boundary pencil, with the endpoint conditions included either in its domain or as finite-dimensional components of an augmented range. At a simple resonance choose right and left states

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad L_n = L(\omega_n),$$

and define, using the bilinear duality of the augmented boundary problem,

$$\alpha_n = \left\langle \widetilde{u}_n, L'(\omega_n) u_n \right\rangle, \qquad \alpha_n \neq 0.$$

Let the repeated block be

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix}, \qquad \beta_n = \langle \widetilde{u}_n, K(\omega_n) u_n \rangle.$$

All endpoint components of the augmented pencil and its duality are included in these pairings.

Theorem 11.1 (Resonant evaluation theorem)**.** *In endpoint frames compatible with the partial jet,*

$$\boxed{\kappa_n := \frac{b(\omega_n)}{a'(\omega_n)} = \frac{\beta_n}{\alpha_n} = -\omega_n'(0).}$$

Here $\omega_n(\tau)$ is the resonance branch of

$$L(\omega, \tau) = L(\omega) + \tau K(\omega)$$

through ω_n. The equality is invariant under a common analytic nonzero Evans unit.

Proof. Differentiate

$$L(\omega_n(\tau), \tau)u(\tau) = 0$$

at $\tau = 0$:

$$L_n\dot{u} + \omega_n'(0)L_n'u_n + K_nu_n = 0.$$

Pairing with $\widetilde{u}_n$ removes the first term and gives

$$\omega_n'(0)\alpha_n + \beta_n = 0.$$

On the connection side, differentiation of

$$a(\omega_n(\tau), \tau) = 0$$

gives

$$a'(\omega_n)\omega_n'(0) + \partial_\tau a(\omega_n, 0) = 0.$$

The exact partial-jet construction identifies $\partial_\tau a(\omega_n, 0) = b(\omega_n)$. Multiplication of the Evans coefficient by an analytic unit $u(\omega, \tau)$ multiplies both a' and b at the root by $u(\omega_n, 0)$, so the ratio is unchanged. □

Proposition 11.2 (Resonant functional on the extension class)**.** *The assignment*

$$\mathfrak{M}_n([K]) := \langle \widetilde{u}_n, K(\omega_n)u_n \rangle$$

is a well-defined linear functional on the local self-extension class. Thus nonsplitting is the statement $[K] \neq 0$, while defectiveness at ω_n is the sharper statement $\mathfrak{M}_n([K]) \neq 0$.

Proof. Under a triangular change of splitting,

$$K \longmapsto K + LQ - QL.$$

At the resonance,

$$\langle \widetilde{u}_n, (L_nQ - QL_n)u_n \rangle = \langle L_n^*\widetilde{u}_n, Qu_n \rangle - \langle \widetilde{u}_n, QL_nu_n \rangle = 0.$$

When endpoint conditions are encoded in an augmented pencil, the same calculation includes the corresponding boundary terms by definition. □

The theorem and proposition separate a global module property from its evaluation at one resonance. A nonzero extension class need not be defective at every QNM: the linear functional $\mathfrak{M}_n$ may vanish at isolated modes.

12 The generalized root vector is non-Einstein

Proposition 12.1 (Nonzero carrier in the generalized root)**.** *Assume*

$$a(\omega_n) = 0, \qquad a'(\omega_n) \neq 0, \qquad b(\omega_n) \neq 0.$$

The geometric root vector belongs to the source Einstein submodule, while every length-two root polynomial has first generalized coefficient with nonzero projection to the carrier quotient. In the normalized triangular frame that projection is

$$-\frac{a'(\omega_n)}{b(\omega_n)} = -\frac{\alpha_n}{\beta_n} = -\frac{1}{\kappa_n}.$$

Thus the defective doubled resonance is not two copies of an Einstein quasinormal mode: its generalized member is intrinsically non-Einstein.

Proof. Put $z = \omega - \omega_n$ and write

$$a = a_1 z + O(z^2), \qquad b = b_0 + O(z), \qquad a_1 b_0 \neq 0.$$

For

$$c(z) = c_0 + z c_1 + O(z^2), \qquad T_2(z)c(z) = O(z^2),$$

the order-zero equation gives $c_0 = e_X$, the source Einstein vector. Write $c_1 = x e_X + y e_Y$. The e_X component at order z is

$$a_1 + b_0 y = 0,$$

so $y = -a_1/b_0 \neq 0$. By Theorem 3.1, e_Y projects nontrivially to the Ricci-carrier quotient. Adding a multiple of e_X changes x, but cannot remove y. $\square$

13 Exterior cut-off outgoing Green double pole

Fix ordinary radii $2 < r_H < r_I < \infty$ inside the certified analytic continuation domains. Let

$$H(\omega) : \mathbb{C}^3 \longrightarrow C^\infty([r_H, r_I]; \mathbb{C}^6)$$

be the analytic Poisson operator for horizon-selected homogeneous data, let $\Gamma_-(\omega)$ be the unwanted incoming trace at r_I, and set

$$T(\omega) = \Gamma_-(\omega) H(\omega).$$

For a compactly supported source F, let $P_H(\omega)F$ denote the inhomogeneous solution with zero horizon data. The outgoing solution is

$$R_{\text{out}}(\omega)F = P_H(\omega)F - H(\omega)T(\omega)^{-1}\Gamma_-(\omega)P_H(\omega)F. \tag{3}$$

Theorem 13.1 (Finite-interval outgoing Green pole)**.** *On a neighborhood of the certified frequency ω_n, the operator*

$$R_{\text{out}}(\omega) : C_c^\infty((r_H, r_I); \mathbb{C}^6) \longrightarrow C_{\text{loc}}^\infty((r_H, r_I); \mathbb{C}^6)$$

is meromorphic and has a nonzero rank-one pole of order two. Up to the endpoint trace sign convention, its principal coefficient is

$$\operatorname{Coeff}_{(\omega-\omega_n)^{-2}} R_{\text{out}} = \frac{b(\omega_n)}{a'(\omega_n)^2} H(\omega_n) e_X \otimes \left[e_Y^* \Gamma_-(\omega_n) P_H(\omega_n) \right].$$

Proof. All factors in (3) except T^{-1} are analytic. After analytic elimination of the spin-one unit, the singular block is

$$T_2^{-1} = \begin{pmatrix} a^{-1} & -ba^{-2} \\ 0 & a^{-1} \end{pmatrix},$$

and therefore

$$\mathrm{Coeff}_{z^{-2}}\, T_2^{-1} = -\frac{b(\omega_n)}{a'(\omega_n)^2} e_X \otimes e_Y^*.$$

The minus sign in (3) gives the displayed convention.

The range vector is nonzero because the homogeneous Poisson map is injective, so $H(\omega_n)e_X \neq 0$. The covector is nonzero by a cutoff Jost construction. Given any incoming coefficient q, choose the corresponding incoming Jost solution $J_- q$ near r_I, multiply it by a smooth cutoff χ that vanishes near r_H and equals one near r_I, and set

$$F = (D - \mathbb{A}(\omega_n))(\chi J_- q).$$

Then F is compactly supported, $P_H(\omega_n)F = \chi J_- q$, and $\Gamma_- P_H(\omega_n)F = q$. Hence the incoming trace map from compact sources is onto. Taking $q = e_Y$ shows that the covector does not vanish. The tensor product is therefore nonzero and rank one. □

13.1 Exact transparent cuts

Put $x = r_*$. On the certified QNM neighborhood, let

$$u_H(x, \omega), \qquad u_\infty(x, \omega)$$

be analytic spin-two Jost solutions satisfying the future-horizon and outgoing-infinity conditions. The scalar Evans coefficient is, up to an analytic unit,

$$a(\omega) = W(u_H, u_\infty).$$

Let χ_{s} and χ_{o} be nontrivial smooth source and observation cutoffs with compact support. Choose finite cuts

$$x_- < \operatorname{supp} \chi_{\mathrm{s}} \cup \operatorname{supp} \chi_{\mathrm{o}} < x_+.$$

At the two cuts impose the exact transparent boundary conditions

$$W(u, u_H)|_{x_-} = 0, \qquad W(u, u_\infty)|_{x_+} = 0.$$

Their coefficients are analytic in ω. Uniqueness of the exterior Jost problems shows that the resulting finite-interval inverse agrees exactly, between the cutoffs, with the meromorphically continued exterior outgoing inverse.

Theorem 13.2 (Exterior cut-off Green pole)**.** *For any nontrivial compactly supported source and observation cutoffs,*

$$\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega) \chi_{\mathrm{s}}$$

has a genuine nonzero rank-one pole of order two at the certified Schwarzschild QNM. In the augmented Fredholm normalization its principal coefficient is

$$\boxed{\mathrm{Coeff}_{(\omega-\omega_n)^{-2}}\big(\chi_{\mathrm{o}} R_{\mathrm{ext}} \chi_{\mathrm{s}}\big) = -\frac{\beta_n}{\alpha_n^2} (\chi_{\mathrm{o}} u_n) \otimes (\widetilde{u}_n \chi_{\mathrm{s}}).}$$

Proof. The transparent conditions select exactly the restrictions of the exterior horizon and infinity Jost solutions. Hence

$$\chi_{\mathrm{o}} R_{\mathrm{ext}}(\omega)\chi_{\mathrm{s}} = \chi_{\mathrm{o}} R_{[x_-,x_+]}(\omega)\chi_{\mathrm{s}}.$$

Theorem 13.1 and (4) give the displayed principal coefficient. It is nonzero: $\beta_n \neq 0$, while a nonzero solution of a second-order ODE cannot vanish on a nonempty open interval. Thus $\chi_{\mathrm{o}} u_n$ and $\widetilde{u}_n \chi_{\mathrm{s}}$ are nonzero for every nontrivial smooth cutoff. $\square$

Corollary 13.3 (Non-annihilation by certified reconstruction)**.** *The certified rational reconstruction into the axial metric/curvature variables does not annihilate the range $H(\omega_n)e_X$. Hence at least one reconstructed local observable has the same order-two radial response pole.*

In the finite-interval Fredholm normalization of Section 11, the scalar resolvent has

$$L(\omega)^{-1} = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n(\omega - \omega_n)} + O(1).$$

The upper-right block of $\mathbb{L}^{-1}$ is $-L^{-1}KL^{-1}$, and Theorem 11.1 therefore gives the invariant principal coefficient

$$\boxed{\mathrm{Coeff}_{(\omega-\omega_n)^{-2}}(\mathbb{L}^{-1})_{12} = -\frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n = -\frac{\kappa_n}{\alpha_n} u_n \otimes \widetilde{u}_n.} \tag{4}$$

The Poisson/trace coefficient in Theorem 13.1 and the exterior coefficient in Theorem 13.2 are the same rank-one residue expressed in endpoint and augmented Fredholm coordinates.

14 Conditional global Fredholm promotion

Theorem 13.2 is an exterior differential Green-operator theorem after compact source and observation localization. A global physical resolvent without cutoffs requires a separate closed-operator realization. The correct abstract block notation is

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

where $L(\omega) : D \subset X \to Y$ is an analytic Fredholm pencil of index zero and $K(\omega) : D \to Y$ is graph-norm bounded. Suppose

$$u_n \in \ker L(\omega_n), \qquad \widetilde{u}_n \in \ker L(\omega_n)^*,$$

and define using the Y^*-Y bilinear duality

$$\alpha_n = \langle \widetilde{u}_n, L'(\omega_n)u_n \rangle, \qquad \beta_n = \langle \widetilde{u}_n, K(\omega_n)u_n \rangle.$$

Proposition 14.1 (Conditional global pole formula)**.** *If L^{-1} is meromorphic near ω_n, its zero is simple, $\alpha_n \neq 0$, and $\beta_n \neq 0$, then*

$$\mathbb{L}^{-1} = \begin{pmatrix} L^{-1} & -L^{-1}KL^{-1} \\ 0 & L^{-1} \end{pmatrix}$$

has an order-two pole with principal coefficient

$$-\frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n$$

in the upper-right block.

This proposition is standard block algebra once the Fredholm spaces and pairing exist. What remains open is not the exterior cut-off pole proved above, but the construction of:

1. closed weighted horizon and null-infinity domains;
2. a physical gauge quotient and bounded metric reconstruction;
3. a causal exterior resolvent on a Laplace contour;
4. a justified contour deformation and control of the non-pole part.

Only after these steps could the double pole be promoted to a rigorous $te^{i\omega_n t}$ contribution to retarded ringdown.

15 Critical Einstein–Weyl mass comparison

On the transverse-traceless spin-two sector, the quadratic Einstein–Weyl parent equation is

$$(\mathcal{E} + m\mathcal{A})\phi = 0,$$

where the parameter m has the dimensions of squared mass. Its Schwarzschild axial scalar reduction is

$$L_m(\omega)y = \left[D^2 + \omega^2 - V_2 - mf\right] y = 0, \qquad f = 1 - \frac{2}{r}. \tag{5}$$

For a differentiable family $y(m)$, differentiation at $m = 0$ gives

$$L_2\, \partial_m y = f\, y.$$

Theorem 15.1 (Exact local critical-mass-jet identification)**.** *The projective cocycle of the transverse-traceless critical mass jet is*

$$[\mathcal{I}_{\text{mass}}] = [f].$$

Consequently

$$\boxed{[\mathcal{I}_{\text{Bach}}] = \frac{i\omega}{2}[\mathcal{I}_{\text{mass}}].}$$

With the convention $L_m = L_2 - mf$, the local first-order parameter relation is

$$\boxed{m = \frac{i\omega}{2}\tau}$$

modulo rational triangular gauge.

Proof. Equation (5) gives the extension equation $L_2\partial_m y = fy$, so its scalar projective density is f. Theorem 6.1 gives $[\mathcal{I}_{\text{Bach}}] = (i\omega/2)[f]$. Comparing the two tangent equations yields the parameter relation. □

This theorem closes the local transverse-traceless differential-module comparison. We next show that the same gauge matches the differentiated physical Jost classes.

15.1 Endpoint-compatible mass gauge

The rational representative in Theorem 6.1 satisfies

$$q(r,\omega) = -\frac{i}{8\omega}r + O(1), \qquad r \to \infty.$$

Proposition 15.2 (Forced moving-phase gauge)**.** *Every rational q satisfying*

$$\mathcal{I}_{\mathrm{Bach}} = \mathcal{K}_{U_2}q + \frac{i\omega}{2}f$$

has the forced leading behavior

$$q(r,\omega) = -\frac{i}{8\omega}r + O(1).$$

Thus the equivalence gauge is necessarily unbounded in a fixed massless phase frame.

Proof. At infinity,

$$\mathcal{I}_{\mathrm{Bach}} - \frac{i\omega}{2}f \longrightarrow -\frac{i\omega}{2}, \qquad U_2 \longrightarrow \omega^2.$$

A rational $q = o(r)$ is $O(1)$ and gives $\mathcal{K}_{U_2}q = o(1)$, so it cannot produce the nonzero constant. If $q \sim cr$, then $Dq \to c$, while $D^3q \to 0$ and $D(U_2)q \to 0$. Therefore

$$\mathcal{K}_{U_2}q \longrightarrow 4\omega^2 c.$$

Matching the constant gives $c = -i/(8\omega)$, uniquely. □

Factor normalization. The normalization in Lemma 5.1 must be retained. Define the field-redefinition operator

$$\boxed{Q_q = 2qD - D(q).}$$

Then

$$[L_2, Q_q]y = -\mathcal{K}_{U_2}(q)y, \qquad L_2y = 0.$$

Consequently, if

$$L_2x = \mathcal{I}_{\mathrm{Bach}}y, \qquad \widehat{x} = x + Q_qy,$$

then

$$\boxed{L_2\widehat{x} = \frac{i\omega}{2}fy.} \tag{6}$$

The factor 2 in Q_q is essential.

Theorem 15.3 (Coulomb cancellation and differentiated Jost classes)**.** *Let $y_\sigma(m,r)$, $\sigma = \pm 1$, be a massive spin-two Jost germ at infinity, with $k = (\omega^2 - m)^{1/2}$ on the branch through $k(0) = \omega$. It has the form*

$$y_\sigma(m,r) = e^{\sigma ikr}r^{\rho_\sigma(m)}\left(1 + O(r^{-1})\right), \qquad \rho_\sigma(m) = \sigma i\left(2k + \frac{m}{k}\right).$$

In particular,

$$\rho'_\sigma(0) = 0, \qquad \frac{\partial_m y_\sigma(0,r)}{y_\sigma(0,r)} = -\frac{\sigma i}{2\omega}r + O(1).$$

Writing $c(\omega) = i\omega/2$, the rational gauge satisfies

$$\frac{Q_q y_\sigma}{y_\sigma} = \frac{\sigma r}{4} + O(1) = c(\omega)\frac{\partial_m y_\sigma(0,r)}{y_\sigma(0,r)} + O(1).$$

At the horizon the mass perturbation is multiplied by $f = O(r-2)$; the indicial exponent is unchanged, q is regular, and Q_q preserves the phase-factored analytic ingoing germ. Hence, at either endpoint $e \in \{\mathcal{H}^+, \mathcal{I}^+\}$,

$$\boxed{\widehat{x}_e - c(\omega)\partial_m y_e(0) = \lambda_e(\omega) y_e}$$

for an analytic scalar λ_e.

Proof. At infinity the massive equation is

$$D^2 y + \left(k^2 + \frac{2m}{r} + O(r^{-2})\right) y = 0.$$

Substitution of the displayed ansatz at order r^{-1} gives

$$2\sigma i k \rho_\sigma + 4k^2 + 2m = 0,$$

and therefore the stated ρ_σ. Since

$$k'(0) = -\frac{1}{2\omega}, \qquad \frac{d}{dm}\left(2k + \frac{m}{k}\right)\bigg|_{m=0} = -\frac{1}{\omega} + \frac{1}{\omega} = 0,$$

the logarithmic derivative cancels. The remaining phase derivative gives the linear term in $\partial_m y_\sigma / y_\sigma$. Using $q = -ir/(8\omega) + O(1)$ and $Dy_\sigma/y_\sigma = \sigma i\omega + O(r^{-1})$ gives

$$2q\frac{Dy_\sigma}{y_\sigma} - D(q) = \frac{\sigma r}{4} + O(1).$$

Equation (6) shows that $\widehat{x} - c\,\partial_m y$ is homogeneous. The asymptotic calculation places it in the endpoint Jost line, proving the infinity statement. Regular Frobenius dependence and the unchanged indicial exponent prove the horizon statement. $\square$

Theorem 15.4 (Critical-mass Evans derivative and QNM velocity)**.** *Let $a(\omega, m)$ be the massive spin-two Evans coefficient in analytic horizon and infinity Jost normalizations, and let $b_{\mathrm{B}}(\omega)$ be the Bach connection shear. There is an analytic scalar $h(\omega)$ such that*

$$\boxed{b_{\mathrm{B}}(\omega) = \frac{i\omega}{2}\,\partial_m a(\omega, 0) + h(\omega) a(\omega, 0).}$$

At the certified simple QNM,

$$b_{\mathrm{B}}(\omega_n) = \frac{i\omega_n}{2}\partial_m a(\omega_n, 0).$$

The massive QNM branch supplied by the implicit-function theorem therefore satisfies

$$\boxed{\frac{d\omega_n}{dm}\bigg|_{m=0} = \frac{2i}{\omega_n}\kappa_n \neq 0.} \tag{7}$$

Proof. The endpoint corrections in Theorem 15.3 are scalar changes of the differentiated Jost normalizations. Their contribution to the differentiated connection coefficient is therefore an analytic multiple $h(\omega)a(\omega,0)$. At a zero of a that term vanishes. Differentiating $a(\omega_n(m),m)=0$ and using $\kappa_n = b_{\mathrm{B}}(\omega_n)/a'(\omega_n)$ gives

$$\kappa_n = -\frac{i\omega_n}{2}\,\frac{d\omega_n}{dm}\bigg|_{m=0}.$$

The displayed formula follows, and it is nonzero because the certified selector satisfies $\kappa_n \neq 0$. ☐

Corollary 15.5 (Unified resonance invariant)**.** *At the certified mode,*

$$\boxed{\kappa_n = \frac{b_n}{a'_n} = \frac{\beta_n}{\alpha_n} = -\frac{i\omega_n}{2}\,\frac{d\omega_n}{dm}\bigg|_0}\,.$$

Consequently

$$(c_1)_Y = -\frac{1}{\kappa_n} = \frac{2}{i\omega_n\,\omega'_n(0)}$$

and, for the upper-right triangular Green block,

$$R^{(12)}_{-2} = -\frac{\kappa_n}{\alpha_n}u_n \otimes \widetilde{u}_n = \frac{i\omega_n}{2\alpha_n}\omega'_n(0)u_n \otimes \widetilde{u}_n.$$

Corollary 15.6 (Reflected EP2 pair)**.** *With the $e^{i\omega t}$ convention, the real Schwarzschild coefficients and QNM endpoint conditions obey*

$$L(-\overline{\omega},m) = \overline{L(\omega,m)}.$$

Hence the certified defective resonance near

$$\omega_n = -0.3736716844180418 + 0.0889623156889357\,i$$

has a reflected defective partner near

$$\boxed{\omega_n^\sharp = +0.3736716844180418 + 0.0889623156889357\,i.}$$

Their velocities and projective selectors satisfy

$$\nu_n^\sharp = -\overline{\nu_n}, \qquad \kappa_n^\sharp = -\overline{\kappa_n}.$$

In particular, nonvanishing of the certified selector supplies a symmetry-related pair of EP2s. If P_n, C_{-2}, and C_{-1} denote respectively the ordinary residue and the two singular coefficients of the critical response, then

$$P_n^\sharp = -\overline{P_n}, \qquad C_{-2}^\sharp = \overline{C_{-2}}, \qquad C_{-1}^\sharp = -\overline{C_{-1}}.$$

The even/odd signs follow from the pole order under $\omega \mapsto -\overline{\omega}$, and allow the pair to combine into a real perturbation.

15.2 Boundary transgression as an audit

For finite cuts, the direct field-redefinition convention gives

$$\boxed{\beta^{\mathrm{Bach}}_{n,[x_-,x_+]} - \frac{i\omega_n}{2}\beta^{\mathrm{mass}}_{n,[x_-,x_+]} = -\left[W(\widetilde{u}_n, Q_q u_n)\right]^{x_+}_{x_-}.}$$

In the augmented Jost pencil, the endpoint-map derivatives absorb this term. Theorem 15.4 shows that an explicit evaluation of the individual endpoint constants is unnecessary at the zero divisor: their net effect is the term $h(\omega)a(\omega,0)$.

16 Universal critical resonance and the covariant parent

Theorem 16.1 (Universal critical-resonance criterion). *Let $L(\omega) : D \to Y$ be an analytic Fredholm pencil with fixed domain, and suppose ω_n is a simple resonance. Let*

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad \alpha_n = \langle \widetilde{u}_n, L_n' u_n \rangle \neq 0.$$

For an analytic deformation $L_m = L + mA$, put

$$\beta_n = \langle \widetilde{u}_n, A(\omega_n) u_n \rangle$$

in the augmented endpoint pairing. If $R = L^{-1}$, then

$$C(\omega) = R(\omega)A(\omega)R(\omega)$$

has expansion

$$\boxed{C(\omega) = \frac{\beta_n}{\alpha_n^2}\frac{u_n \otimes \widetilde{u}_n}{(\omega-\omega_n)^2} + \frac{C_{-1}}{\omega-\omega_n} + C_{\mathrm{hol}}(\omega).}$$

Consequently C has an exact pole of order two if and only if $\beta_n \neq 0$; at a simple resonance it cannot have a pole of order three. If $\omega_n(m)$ is the deformed resonance branch, then

$$\boxed{\nu_n := \omega_n'(0) = -\frac{\beta_n}{\alpha_n}, \qquad C_{-2} = -\nu_n P_n, \qquad P_n = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n}.}$$

Proof. Write

$$R(\omega) = \frac{P_n}{\omega-\omega_n} + H_n + O(\omega-\omega_n).$$

The only second-order term in RAR is

$$P_n A_n P_n = \frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n.$$

Differentiating $L_m(\omega_n(m))u_n(m) = 0$ and pairing with $\widetilde{u}_n$ gives $\beta_n + \nu_n\alpha_n = 0$. □

Proposition 16.2 (Canonical simple pole and tangent state). *Let $z = \omega - \omega_n$ and expand*

$$R(\omega) = \frac{P_n}{z} + H_n + O(z), \qquad A(\omega) = A_n + zA_n' + O(z^2).$$

Then the critical response $C = RAR = -\partial_m R_m|_{m=0}$ has

$$\boxed{C_{-2} = P_n A_n P_n = -\nu_n P_n}$$

and

$$\boxed{C_{-1} = P_n A_n H_n + H_n A_n P_n + P_n A_n' P_n = -\dot{P}_n.}$$

The term $P_n A_n' P_n$ is absent only when the deformation operator is frequency independent in the fixed-domain pencil.

Let

$$g_n = (A_n + \nu_n L_n')u_n.$$

Then the right and left tangent classes are

$$\boxed{[\dot{u}_n] = [-H_n g_n] \in D/\mathbb{C}u_n, \qquad [\dot{\widetilde{u}}_n] = [-\widetilde{u}_n(A_n + \nu_n L_n')H_n] \bmod \mathbb{C}\widetilde{u}_n.}$$

Proof. Direct multiplication of the two displayed expansions gives both Laurent coefficients, including the A_n' term. Differentiating the moving simple pole gives $C_{-1} = -\dot{P}_n$.

The differentiated QNM equation is

$$L_n \dot{u}_n + g_n = 0, \qquad \langle \widetilde{u}_n, g_n \rangle = \beta_n + \nu_n \alpha_n = 0.$$

The Laurent identities imply

$$L_n H_n = I - L_n' P_n.$$

Since $P_n g_n = 0$, the vector $-H_n g_n$ solves the equation; the remaining freedom is a multiple of u_n. The left formula follows from the corresponding right Laurent identity. □

The tangent vector $\dot{u}_n$ depends on normalization, but its class

$$\boxed{[\dot{u}_n] \in D/\mathbb{C}u_n}$$

is canonical and obeys

$$L_n \dot{u}_n = -(A_n + \nu_n L_n')u_n.$$

In the axial Bach filtration its carrier projection is $-\alpha_n/\beta_n = -1/\kappa_n$. Thus the generalized QNM is the normalization-independent tangent class of the massive QNM branch. Proposition 16.2 also shows that it can be reconstructed from the scalar QNM, its adjoint, and the finite part of the ordinary Regge–Wheeler resolvent; the six-state integration remains an independent certificate rather than an analytic necessity.

Proposition 16.3 (Augmented QNM Hellmann–Feynman formula)**.** *For a scalar realization*

$$L(\omega, m)y = [\partial_x^2 + Q(x, \omega, m)]y = 0,$$

let y_- and y_+ be the left and right Jost solutions and choose the Evans orientation $a = W(y_-, y_+)$. For any parameter p,

$$\boxed{\partial_p a = \int_{x_-}^{x_+} y_-(x)\, \partial_p Q(x, \omega, m)\, y_+(x)\, dx + B_p,}$$

where B_p is the transparent-boundary contribution (equivalently, the endpoint-normalization term) in the augmented pairing. At a simple QNM,

$$\boxed{\nu_n = -\frac{\partial_m a}{\partial_\omega a} = -\frac{\langle \widetilde{u}_n, A_n u_n \rangle_{\text{aug}}}{\langle \widetilde{u}_n, L_n' u_n \rangle_{\text{aug}}}.}$$

For $Q = \omega^2 - V_2 - mf$, this is

$$\boxed{\nu_n = -\frac{-\displaystyle\int f\, u_n^2\, dx + B_m}{2\omega_n \displaystyle\int u_n^2\, dx + B_\omega}.}$$

The integrals are bilinear QNM integrals, not positive L^2 norms.

Proof. Differentiate the two scalar equations, apply the Lagrange identity, and integrate between the endpoint cuts. The bulk term is the displayed integral; differentiating the analytic transparent maps supplies B_p. At a QNM the two Jost solutions are proportional, and the proportionality cancels from $-a_m/a_\omega$. The augmented formula is the same identity written invariantly. □

On a fixed gauge-reduced domain, let

$$\mathcal{E}_m(\omega) = \mathcal{E}(\omega) + m\mathcal{A}.$$

The metric–metric block of the inverse parent Hessian is

$$G^{\mathrm{W}}_{hh}(\omega) = \frac{1}{4\alpha_{\mathrm{W}}} \mathcal{E}(\omega)^{-1} \mathcal{A} \mathcal{E}(\omega)^{-1}.$$

Theorem 16.4 (Critical mass derivative of the metric Green operator)**.** *Where the inverses exist,*

$$\boxed{G^{\mathrm{W}}_{hh}(\omega) = -\frac{1}{4\alpha_{\mathrm{W}}}\, \partial_m \mathcal{E}_m(\omega)^{-1}\big|_{m=0}.}$$

No commutativity between $\mathcal{E}$ and $\mathcal{A}$ is required. If, on a fixed-domain meromorphic realization,

$$\mathcal{E}_m(\omega)^{-1} = \frac{P_n(m)}{\omega - \omega_n(m)} + H_n(\omega, m),$$

then

$$\boxed{G_{-2} = -\frac{\nu_n}{4\alpha_{\mathrm{W}}} P_n, \qquad G_{-1} = -\frac{1}{4\alpha_{\mathrm{W}}} \dot{P}_n.}$$

Proof. Differentiate $\mathcal{E}_m \mathcal{E}_m^{-1} = I$:

$$\partial_m \mathcal{E}_m^{-1} = -\mathcal{E}_m^{-1} \mathcal{A} \mathcal{E}_m^{-1}.$$

Differentiating the simple-pole expansion gives

$$\partial_m \mathcal{E}_m^{-1}\big|_0 = \frac{\nu_n P_n}{(\omega - \omega_n)^2} + \frac{\dot{P}_n}{\omega - \omega_n} + \dot{H}_n(\omega, 0).$$

□

The corresponding exact finite-mass secant identity is

$$\boxed{\frac{\mathcal{E}^{-1} - \mathcal{E}_m^{-1}}{m} = \mathcal{E}^{-1} \mathcal{A} \mathcal{E}_m^{-1} \longrightarrow \mathcal{E}^{-1} \mathcal{A} \mathcal{E}^{-1}.}$$

It follows from the noncommutative resolvent identity and makes the Weyl metric propagator the confluent difference quotient of the massless and massive second-order propagators.

For

$$P_n = \frac{u_n \otimes \widetilde{u}_n}{\alpha_n}, \qquad \alpha_n = \langle \widetilde{u}_n, \partial_\omega \mathcal{E}(\omega_n) u_n \rangle, \qquad \beta_n = \langle \widetilde{u}_n, \mathcal{A} u_n \rangle,$$

the differentiated eigenvalue equation gives

$$\alpha_n \nu_n + \beta_n = 0.$$

Hence

$$G_{-2} = \frac{1}{4\alpha_{\mathrm{W}}} \frac{\beta_n}{\alpha_n^2} u_n \otimes \widetilde{u}_n = -\frac{i}{2\alpha_{\mathrm{W}}\omega_n} \kappa_n P_n.$$

Thus parent block inversion, mass differentiation, the resonant overlap, and the projective selector give the same nonzero double coefficient. This metric–metric block should not be confused with the full triangular root-space coefficient in Theorem 17.3: the latter is nilpotent under composition on the two-layer extension space. The projected rank-one coefficient C_{-2} or G_{-2} need not itself square to zero on the projected source/response space.

Theorem 16.5 (Isolated parent-resonance contribution)**.** *For a small contour enclosing only the massive QNM branch,*

$$\boxed{\mathcal{U}_n^{\mathrm{W}}(t) = -\frac{e^{i\omega_n t}}{4\alpha_{\mathrm{W}}} \left[\dot{P}_n + it\,\nu_n P_n\right].}$$

For a compact source F and an observation functional Λ, the coefficient of $te^{i\omega_n t}$ is

$$\boxed{A_t(F,\Lambda) = -\frac{i\nu_n}{4\alpha_{\mathrm{W}}\alpha_n} \Lambda(u_n) \langle \widetilde{u}_n, F\rangle.}$$

It is present exactly when the source excites the adjoint resonant channel and the observable sees the Einstein QNM profile.

Proof. Differentiate the ordinary isolated-pole contribution $e^{i\omega_n(m)t}P_n(m)$ and apply Theorem 16.4. The second formula follows from the rank-one expression for P_n. $\square$

The polynomial term is Einstein-shaped even though the generalized vector has a nonzero carrier quotient:

$$e^{i\omega_n t}(V_1 + itV_0).$$

The simple-pole coefficient contains the deformation of the mode shape and normalization. Explicitly,

$$\dot{P}_n = \frac{\dot{u}_n \otimes \widetilde{u}_n + u_n \otimes \dot{\widetilde{u}}_n}{\alpha_n} - \frac{\dot{\alpha}_n}{\alpha_n} P_n.$$

As before, this is an isolated local contour theorem, not a global retarded contour deformation.

Corollary 16.6 (Higher critical jets)**.** *For $p \geq 1$, define*

$$C_p = R(AR)^p.$$

Then

$$\boxed{C_p = \frac{(-1)^p}{p!}\, \partial_m^p (L + mA)^{-1}\big|_{m=0}.}$$

If $\beta_n \neq 0$, its pole order at ω_n is $p+1$, with leading coefficient

$$\frac{\beta_n^p}{\alpha_n^{p+1}} u_n \otimes \widetilde{u}_n.$$

The leading isolated-contour polynomial is

$$\frac{(it)^p}{p!} e^{i\omega_n t} \frac{\beta_n^p}{\alpha_n^{p+1}} u_n \otimes \widetilde{u}_n.$$

Pure Weyl gravity is the first-jet case $p = 1$.

For further Schwarzschild overtones, the universal criterion replaces a full six-state Smith calculation by two validated scalar tasks: certify a simple Regge–Wheeler QNM and enclose the augmented overlap $\beta_n = \langle \widetilde{u}_n, A_n u_n \rangle$ away from zero. The Hellmann–Feynman identity

$$\omega_n'(0) = -\frac{\langle \widetilde{u}_n, A_n u_n \rangle_{\mathrm{aug}}}{\langle \widetilde{u}_n, L_n' u_n \rangle_{\mathrm{aug}}}$$

then supplies the mass velocity, pole order, and generalized root length. Proposition 16.3 reduces the overlap to a bilinear scalar Wronskian integral plus explicit endpoint terms. A validated evaluation for several overtones would decide whether the certified reflected EP2 pair is the first member of a defective tower; no such tower is claimed here.

17 Transverse unfolding of the defective resonance

Put

$$\nu_n = \left.\frac{d\omega_n}{dm}\right|_{m=0} = \frac{2i}{\omega_n} \kappa_n \neq 0.$$

The massless divisor remains at ω_n, while the massive divisor is

$$\omega_m = \omega_n + \nu_n m + O(m^2).$$

Proposition 17.1 (Generic versus filtration-preserving splitting)**.** *Put the critical block in the form*

$$T_0(z) = \begin{pmatrix} z & -1 \\ 0 & z \end{pmatrix}.$$

For a general first-order perturbation,

$$T(z, m) = T_0(z) + m \begin{pmatrix} a & b \\ c & d \end{pmatrix} + O(z^2, mz, m^2),$$

one has

$$\det T = z^2 + m(a + d)z + mc + O(z^3, mz^2, m^2).$$

If $c \neq 0$, the two roots have the Newton–Puiseux expansion

$$z_\pm(m) = \pm\sqrt{-cm} - \frac{a + d}{2} m + O(m^{3/2}).$$

If the perturbation preserves the Einstein submodule, then $c = 0$ identically and the two roots are analytic and split linearly. The Einstein–Weyl mass deformation is of this filtered type: the Einstein branch remains massless while the carrier branch obeys the massive equation.

For a generic square-root unfolding, the branch separation is $O(|m|^{1/2})$, the projectors are $O(|m|^{-1/2})$, and a positive branch metric typically has condition number $O(|m|^{-1})$. The filtration-protected linear splitting below instead produces the stronger exponents $O(|m|^{-1})$ and $O(|m|^{-2})$.

Theorem 17.2 (Filtered critical-mass unfolding normal form)**.** *Near $(\omega_n, 0)$, the two-channel connection pencil is analytically left–right equivalent to*

$$\boxed{\mathcal{T}(z,\mu) = \begin{pmatrix} z & -1 \\ 0 & z-\mu \end{pmatrix},}$$

where $z = z(\omega, m)$ and $\mu = \mu(m)$ are analytic local coordinates,

$$z(\omega_n, 0) = 0, \qquad \partial_\omega z(\omega_n, 0) \neq 0,$$

and

$$\mu(m) = \partial_\omega z(\omega_n, 0)\, \nu_n m + O(m^2).$$

Thus the pure-Weyl EP2 is a filtration-preserving transverse confluence of the massless and massive QNM divisors, not a generic miniversal perturbation of a Jordan block.

Proof. In a finite-mass branch frame the two diagonal divisors are scalar Evans functions $a_0(\omega)$ and $a_m(\omega, m)$. Both have a simple zero at $(\omega_n, 0)$, while Theorem 15.4 says that their zero loci have distinct nonzero tangents in the mass direction. Take z to be an analytic unit multiple of a_0, and take μ to be the normalized difference of the two divisors. The off-diagonal entry in the confluent frame is a unit because the certified Smith type is $(0, 0, 2)$. Analytic invertible row and column operations normalize that unit to -1 and remove the remaining divisible entries. □

Theorem 17.3 (Root-space polarization and nilpotent pole)**.** *Let*

$$\mathbb{L}(\omega) = \begin{pmatrix} L(\omega) & K(\omega) \\ 0 & L(\omega) \end{pmatrix},$$

with

$$L_n u_n = 0, \qquad L_n^* \widetilde{u}_n = 0, \qquad \alpha_n = \langle \widetilde{u}_n, L_n' u_n \rangle \neq 0, \qquad \beta_n = \langle \widetilde{u}_n, K_n u_n \rangle \neq 0.$$

The right and left geometric root vectors are

$$\boxed{V_0 = (u_n, 0), \qquad W_0 = (0, \widetilde{u}_n).}$$

They are self-orthogonal in the full-pencil duality:

$$\langle W_0, V_0 \rangle = 0, \qquad \langle W_0, \mathbb{L}_n' V_0 \rangle = 0.$$

The first generalized right vector has carrier component

$$-\frac{\alpha_n}{\beta_n} u_n = -\frac{1}{\kappa_n} u_n.$$

A left generalized vector can be chosen with Einstein-dual component $-(\alpha_n/\beta_n)\widetilde{u}_n$; both chains cross the filtration at first generalized order. The second-order Laurent coefficient is

$$\boxed{R_{-2} = -\frac{\beta_n}{\alpha_n^2} V_0 \otimes W_0, \qquad R_{-2}^2 = 0.}$$

Its range is the Einstein resonant line and that line lies in its kernel. Consequently R_{-2} is, up to its nonzero scalar normalization, the nilpotent generator N of the dual-number commutant.

Proof. If $(x, cu_n) \in \ker \mathbb{L}_n$, then $L_n x + cK_n u_n = 0$. Pairing with $\widetilde{u}_n$ gives $c\beta_n = 0$, hence $c = 0$. The adjoint argument gives the stated left root. For a right chain $\mathbb{L}_n V_1 + \mathbb{L}_n' V_0 = 0$, write the carrier component of V_1 as γu_n. The scalar solvability condition is $\gamma\beta_n + \alpha_n = 0$. The Laurent coefficient follows from the block inverse. Its square vanishes because $(V_0 \otimes W_0)^2 = \langle W_0, V_0\rangle V_0 \otimes W_0 = 0$. ☐

Corollary 17.4 (Spectral cotangent realization)**.** *On the resonant fiber,*

$$R_{-2} : (B_2/E)^* \longrightarrow E$$

is a nonzero map between one-dimensional spaces and hence an isomorphism, up to the scalar $-\beta_n/\alpha_n^2$. The double pole is the spectral realization of the endpoint cotangent-type duality $B_2/E \simeq E^*$.

The inverse is

$$\mathcal{T}(z,\mu)^{-1} = \begin{pmatrix} z^{-1} & [z(z-\mu)]^{-1} \\ 0 & (z-\mu)^{-1} \end{pmatrix},$$

and

$$\frac{1}{z(z-\mu)} = \frac{1}{\mu}\left(\frac{1}{z-\mu} - \frac{1}{z}\right) \longrightarrow \frac{1}{z^2}. \tag{8}$$

The critical double pole is therefore the confluent difference quotient of the two finite-mass simple poles.

Theorem 17.5 (Confluent projectors and local contour)**.** *Normalize the massive vector by*

$$u_m = u_0 + mv + O(m^2)$$

and use the confluent basis (u_0, v). The branch-to-confluent matrix and the two spectral projectors are

$$S_m = \begin{pmatrix} 1 & 1 \\ 0 & m \end{pmatrix},$$

$$P_0(m) = \begin{pmatrix} 1 & -1/m \\ 0 & 0 \end{pmatrix}, \qquad P_m(m) = \begin{pmatrix} 0 & 1/m \\ 0 & 1 \end{pmatrix}.$$

Hence

$$P_0 + P_m = I, \qquad mP_0 \longrightarrow -N, \qquad mP_m \longrightarrow N, \qquad N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix},$$

and $\|P_0\|, \|P_m\| \asymp |m|^{-1}$. Equivalently, after normalizing the confluent coordinate by the spectral separation $\delta_m = \nu_n m + O(m^2)$, the projector scale is $O(|\delta_m|^{-1}) = O(|\nu_n m|^{-1})$.

For a small contour enclosing only the two resonances, the local meromorphic evolution contribution satisfies

$$\boxed{\mathcal{U}_m(t) \longrightarrow e^{i\omega_n t}\,(I + i\nu_n t N)\,.}$$

Proof. The projector formulas are

$$P_0 = S_m \operatorname{diag}(1,0) S_m^{-1}, \qquad P_m = S_m \operatorname{diag}(0,1) S_m^{-1}.$$

Residue calculus on the local contour gives

$$\mathcal{U}_m(t) = e^{i\omega_n t} P_0(m) + e^{i\omega_m t} P_m(m) = \begin{pmatrix} e^{i\omega_n t} & \dfrac{e^{i\omega_m t} - e^{i\omega_n t}}{m} \\ 0 & e^{i\omega_m t} \end{pmatrix}.$$

Since $\omega_m - \omega_n = \nu_n m + O(m^2)$, the off-diagonal entry tends to $i\nu_n t e^{i\omega_n t}$. ☐

Equation (8) is the divided difference of the simple resolvent. More generally,

$$\frac{F(\omega_n + \nu_n m) - F(\omega_n)}{\nu_n m} \longrightarrow F'(\omega_n)$$

for analytic F. Taking $F(\omega) = e^{i\omega t}$ gives the local Jordan polynomial, while differentiating the massive eigenvector shows

$$V_1 \equiv \partial_m V_m|_{m=0} \qquad (\mathrm{mod}\ \mathbb{C}V_0).$$

Remark 17.6 (Time-domain boundary). Theorem 17.5 proves the Jordan polynomial for the isolated local two-pole contour contribution. It does not supply a global retarded inverse-Laplace deformation or control the non-pole contour.

Remark 17.7 (Higher filtered jets). If $k+1$ filtered branches coalesce linearly and the branch basis is replaced by its confluent divided-difference basis, the limiting connection has a chain of length $k+1$. Its local resolvent has poles through order $k+1$, and the local contour contribution is

$$e^{i\omega_n t} \sum_{j=0}^{k} \frac{(it)^j}{j!} N^j.$$

The present Bach self-extension is the first nontrivial case $k = 1$.

Theorem 17.8 (Confluent branch involution)**.** *Let C_m act by $+1$ on u_0 and by -1 on u_m. In the confluent basis,*

$$C_m = \begin{pmatrix} 1 & -2/m \\ 0 & -1 \end{pmatrix}.$$

It has no finite critical limit, but

$$\boxed{\lim_{m\to 0} mC_m = -2N.}$$

The projectors $(I \pm C_m)/2$ have residues $\mp N$. Since $m = (i\omega/2)\tau$,

$$\boxed{\lim_{\tau\to 0} \tau C_\tau = \frac{4i}{\omega} N.}$$

Thus the unique nilpotent in the dual-number commutant is exactly the renormalized residue of the finite-mass branch-sign observable.

Proof. The equation $C_m(u_0 + mv) = -(u_0 + mv)$ gives $C_m v = -2u_0/m - v$. This proves the matrix and the stated limits. □

Theorem 17.9 (Critical singularity of positive branch metrics)**.** *For real signed $m \neq 0$, suppose a positive branch metric is*

$$H_{\mathrm{br}}(m) = \mathrm{diag}(h_0(m), h_1(m)), \qquad 0 < c \le h_0, h_1 \le C.$$

In the confluent basis,

$$H_m = (S_m^{-1})^\dagger H_{\mathrm{br}} S_m^{-1} = \begin{pmatrix} h_0 & -h_0/m \\ -h_0/m & (h_0 + h_1)/m^2 \end{pmatrix}.$$

Its condition number satisfies

$$\boxed{\mathrm{cond}(H_m) \asymp |m|^{-2}.}$$

In the spectral-separation normalization this is $O(|\nu_n m|^{-2})$. No scalar rescaling of H_m has a bounded, positive-definite, nondegenerate critical limit.

Proof. One has

$$\det H_m = \frac{h_0 h_1}{m^2}, \qquad \lambda_{\max}(H_m) = \frac{h_0 + h_1}{m^2} + O(1),$$

and therefore

$$\lambda_{\min}(H_m) \longrightarrow \frac{h_0 h_1}{h_0 + h_1} > 0.$$

Scaling slower than m^2 leaves the largest eigenvalue unbounded; scaling like m^2 gives a rank-one limit; and scaling faster gives zero. □

Theorem 17.10 (Renormalized Krein limit)**.** *Let $J_{\mathrm{br}} = \operatorname{diag}(1, -1)$. In the confluent basis,*

$$J_m = (S_m^{-1})^\dagger J_{\mathrm{br}} S_m^{-1} = \begin{pmatrix} 1 & -1/m \\ -1/m & 0 \end{pmatrix},$$

so

$$\boxed{mJ_m \longrightarrow \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}.}$$

Moreover

$$H_m = J_m C_m = \begin{pmatrix} 1 & -1/m \\ -1/m & 2/m^2 \end{pmatrix} > 0, \qquad m^2 H_m \longrightarrow \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}.$$

Thus the pure-Weyl hyperbolic plane is the renormalized critical limit of the opposite-sign finite-mass branch form, whereas its positive C-majorant becomes rank-one.

Proposition 17.11 (Canonical pseudospectral scale)**.** *In the normalized chart at $m = 0$,*

$$\left\| \mathcal{T}(z, 0)^{-1} \right\| \asymp |z|^{-2}.$$

Hence the local ε-pseudospectral radius is

$$\boxed{|z| \asymp \sqrt{\varepsilon}}$$

rather than the simple-pole scale $O(\varepsilon)$. For $m \neq 0$, the crossover occurs at $|z| \asymp |\mu| \asymp |\nu_n m|$: outside that scale the pair resembles the critical double pole, while inside it the two simple poles resolve and their projectors have size $O(|\nu_n m|^{-1})$.

Remaining scalar/tensor projection audit

Let R reconstruct the tensor carrier from the scalar master field and let P project the tensor equation back to the scalar equation. The corresponding exact target is

$$K_{\mathrm{radial}} - P\mathcal{A}R = LQ_0 - Q_0 L$$

in the augmented pencil, with its endpoint component recorded explicitly. Section 16 already identifies the tensor parent coefficient through the massive resolvent. Proving this commutator identity would additionally identify every scalar reconstruction normalization and endpoint term coefficientwise; it is no longer needed to establish the nonzero parent double pole.

18 Computational certificates and claim boundary

The computer-assisted inputs are separated from the exact derivations above.

Certificate family	Claim used here	Evidence type
Axial triangular preflight	Exact RW/RW/Maxwell filtration and reconstruction	Exact rational
Projective cocycle	Gauge law, nontrivial extension class, mass normal form	Exact rational
Critical parent mass jet	TT mass variation and difference quotient	Exact covariant/reduced
Analytic continuation	Analytic Jost germs on the declared QNM domain	Exact continuation gate
RW/Maxwell simplicity	Physical-axis simplicity and scalar endomorphisms	Exact rational
Full Evans contour	Zero-free boundary and winding +1	Validated interval
Local selector	$\delta_\tau \neq 0$ on the root disk	Validated interval
Spin-one local unit	Full three-factor Smith type $(0, 0, 2)$	Validated interval
Fredholm promotion	Finite-interval rank-one radial Green pole and transparent-cut input	Exact reduction plus validated inputs

The theorem statements and proofs are the mathematical authority. The machine-readable records pin exact witnesses, interval margins, source hashes, and independently executable checks. A missing input, unsupported domain, timeout, or failed mutation causes rejection rather than partial acceptance.

Explicit non-results. This paper does not establish a causal spacetime resolvent, a complete QNM expansion, a global late-time asymptotic formula, a retarded inverse-Laplace deformation, a time-domain instability, the off-resonance normalization function $h(\omega)$, an all-ℓ Bach nonsplitting theorem, or a quantum positivity statement. The polynomial $te^{i\omega_n t}$ is proved only for the isolated local two-pole contour contribution.

19 Conclusion

The axial Schwarzschild Bach equation contains a repeated Regge–Wheeler factor, but the repetition is not a rational direct sum. It is a non-split self-extension whose projective class is represented by the elementary direction $(i\omega/2)(1 - 2/r)$. That direction is exactly the transverse-traceless critical mass jet. The equivalence gauge has forced linear growth at infinity, but this is the derivative of the moving massive phase rather than an obstruction: the Coulomb tail cancels its apparent logarithmic shear. At the certified damped mode, the scalar zero is simple while the extension selector is nonzero. The resulting defect has concrete differential content: the generalized root has a nonzero Ricci-carrier quotient, the compactly cut-off exterior outgoing Green operator has a genuine rank-one double pole, its full two-layer coefficient is nilpotent, and the massive spin-two QNM moves with nonzero critical slope. At the covariant level, the metric response is the mass derivative of the massive spin-two resolvent: QNM motion controls its double coefficient, while projector motion controls its accompanying simple coefficient.

The physical mass deformation is filtration-preserving rather than a generic square-root unfolding. The protected Einstein branch stays fixed, forcing linear confluence, m^{-1} projector growth, and m^{-2} positive-metric conditioning. The local two-pole contour converges to the Jordan polynomial, the branch-sign residue converges to the unique nilpotent commutant generator, and the renormalized opposite-sign branch form becomes the pure-Weyl hyperbolic plane. A global causal time-domain interpretation still requires the separate retarded contour theory stated above.

References

[1] T. Kapitula, The Evans function and generalized Melnikov integrals, *SIAM Journal on Mathematical Analysis* **30** (1999), 273–297, doi:10.1137/S0036141097327963.